

Pengolahan Data SEM

Eka Nurfani

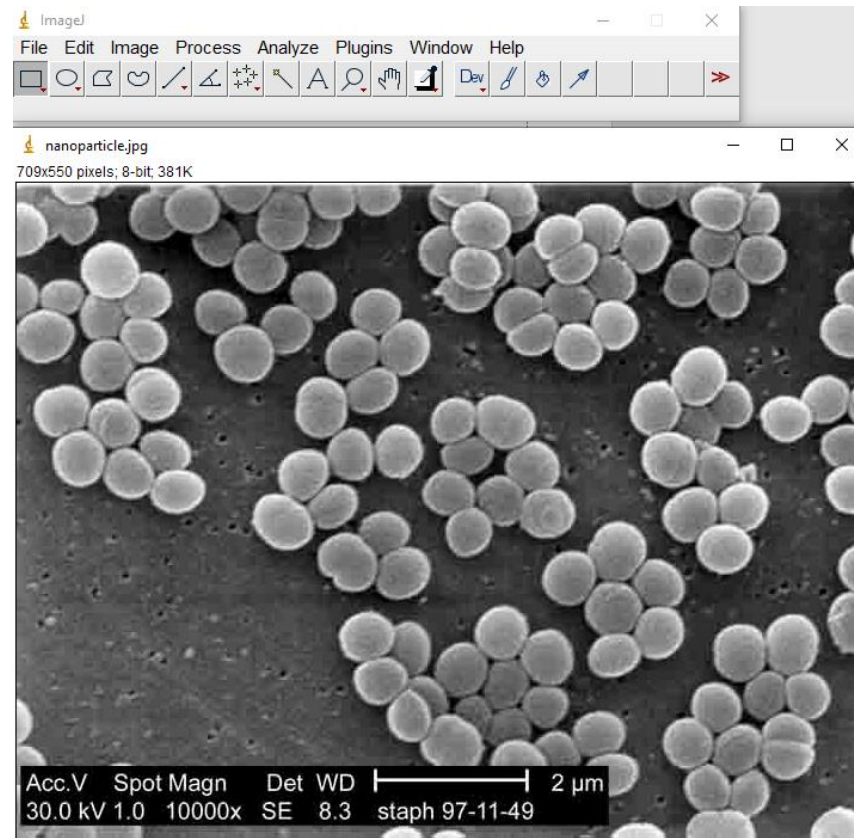
Disampaikan pada **MK Karakterisasi Material Lanjut (MT3015)**

Kelas B

Minggu ke-12

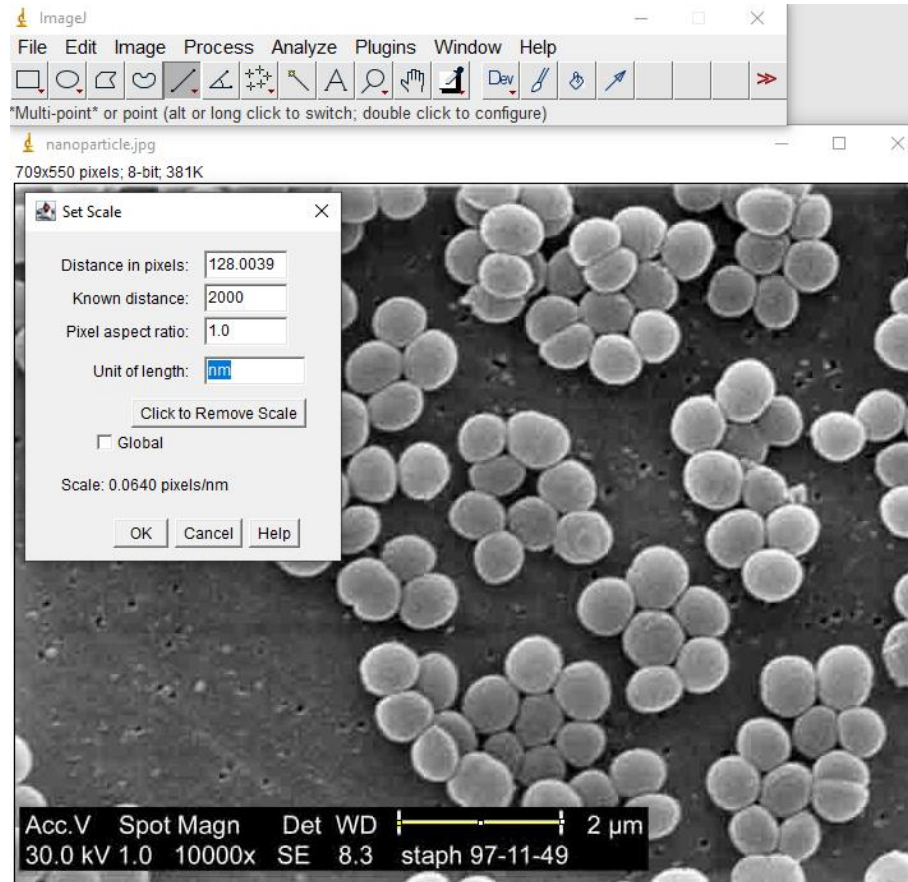
Image-J

- <https://imagej.nih.gov/ij/download.html>
- Download the software
- File >> Open >> Choose image



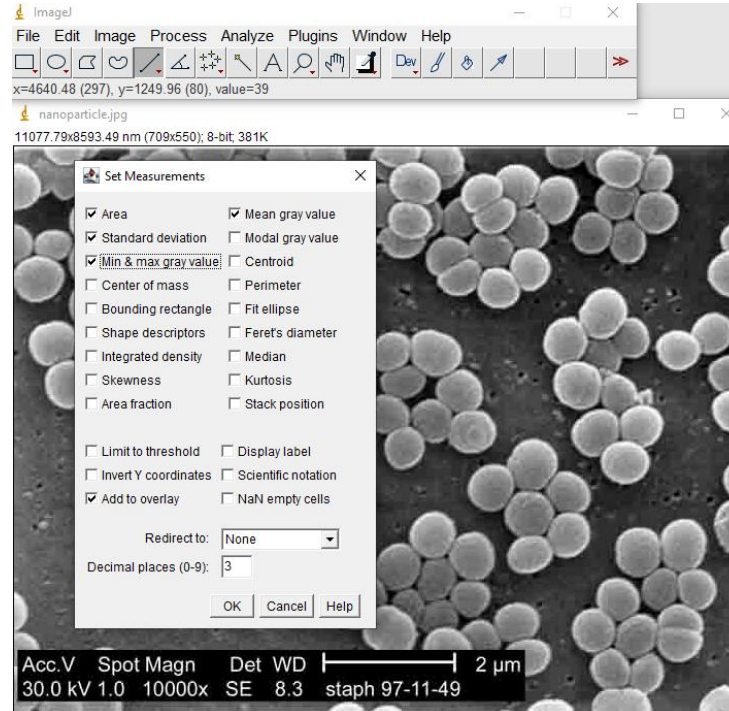
Skala vs Jarak

- Tarik garis lurus pada bar skala
- Analyze >> Set scale



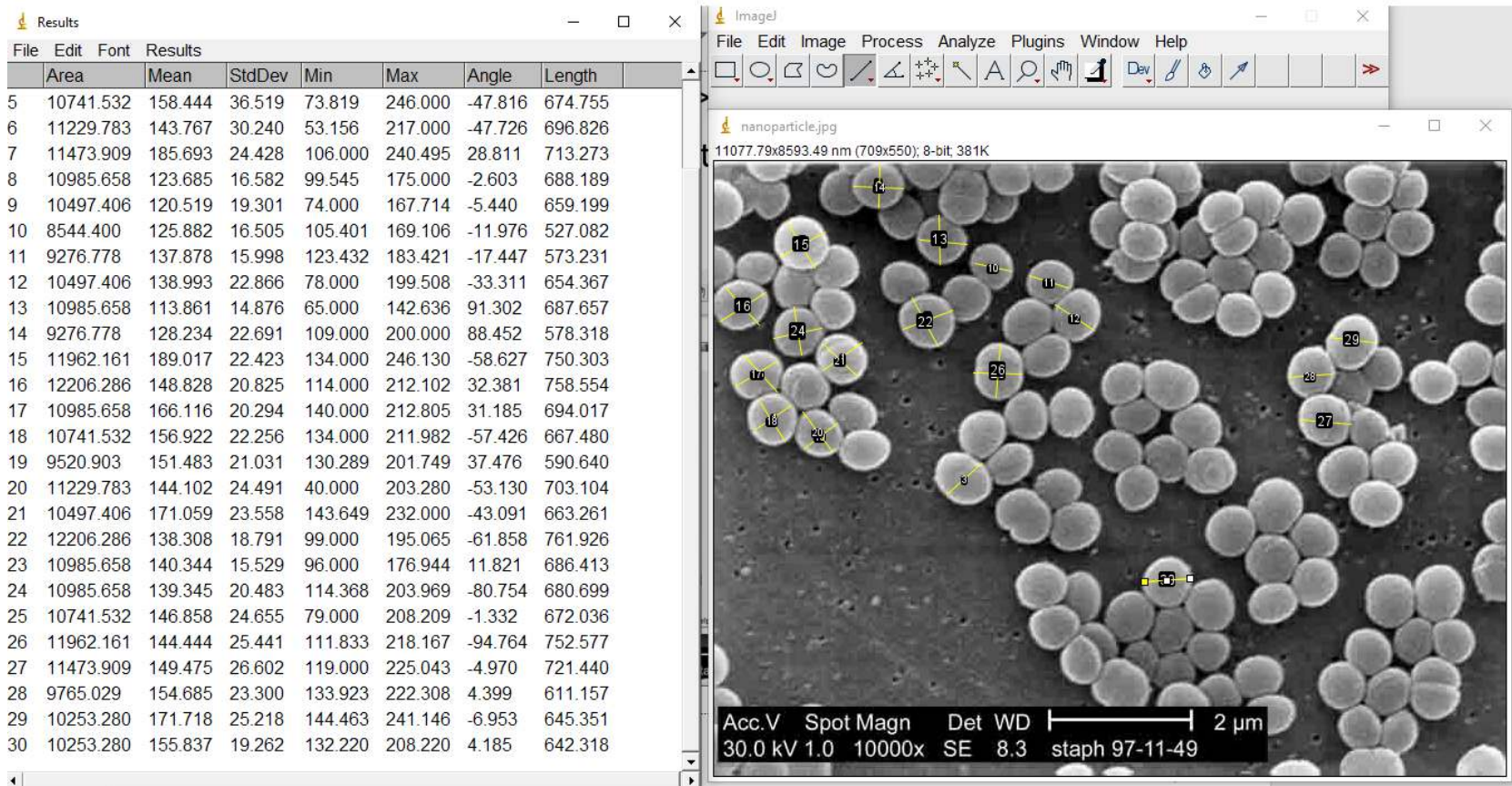
Sampling

- Analyze >> set measurement >> add to overlay
- Tarik garis lurus untuk menghitung Panjang/diameter suatu objek >> tekan M (measure)



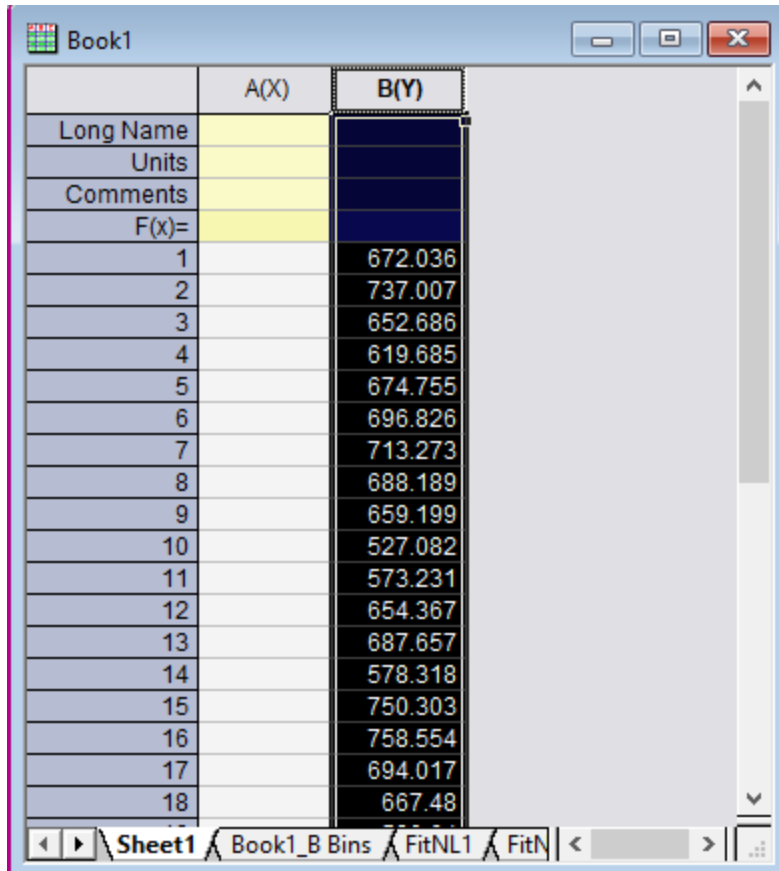
Sampling

- Ukur semua partikel yang terlihat pada gambar
- Simpan hasil Analisa >> Plot dg Excel, Origin, dll.

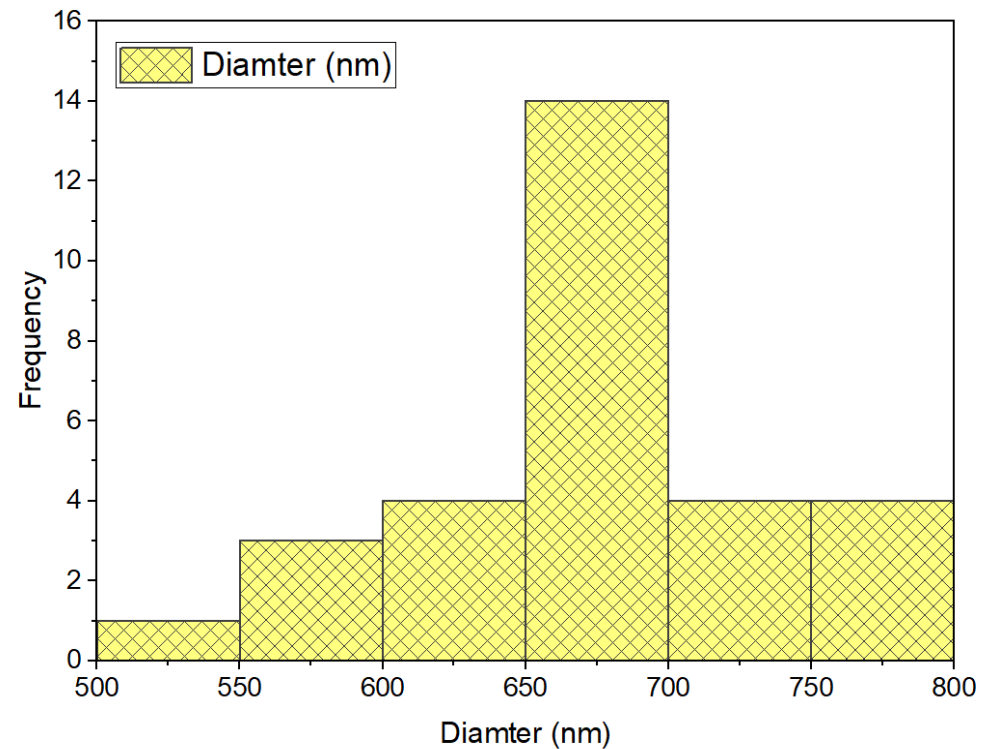


Plot the data

- Plot >> statistic >> histogram



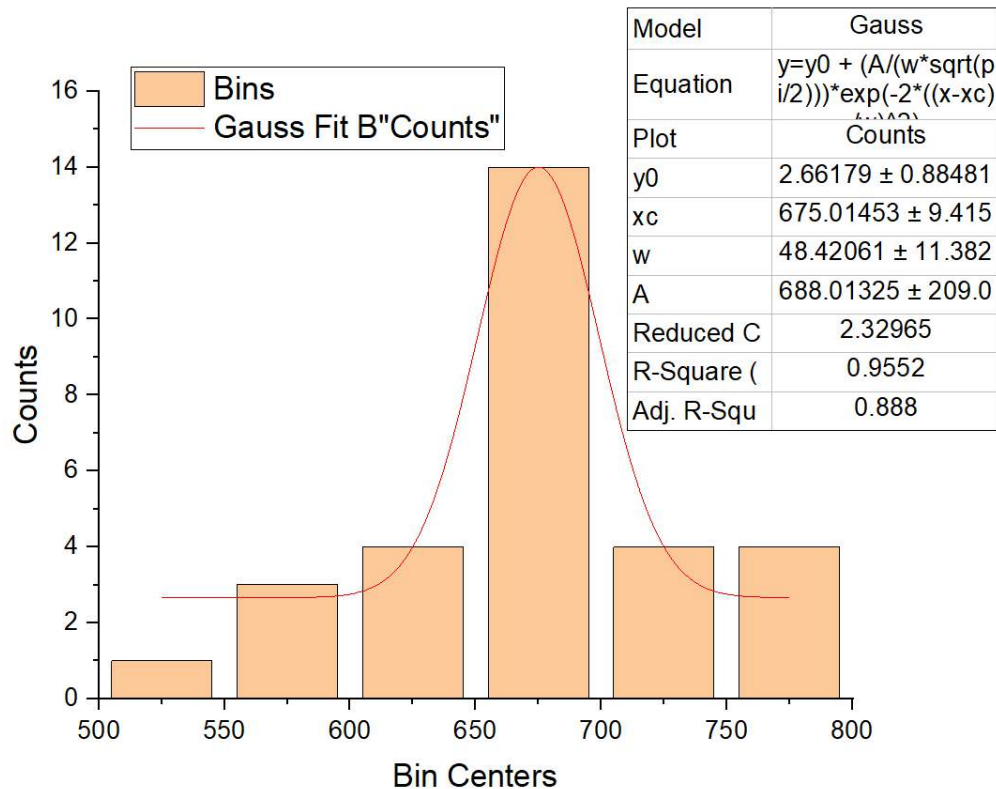
	A(X)	B(Y)
Long Name		
Units		
Comments		
F(x)=		
1		672.036
2		737.007
3		652.686
4		619.685
5		674.755
6		696.826
7		713.273
8		688.189
9		659.199
10		527.082
11		573.231
12		654.367
13		687.657
14		578.318
15		750.303
16		758.554
17		694.017
18		667.48



Analyze the data

- Go to BINS worksheet >> plot column graph >> analysis >> non-linear curve fit

	A(X)	B(Y)	C(Y)	D(Y)
Long Name	Bin Centers	Counts	Cumulative Sum	Cumulative Percent
Units				
Comments	Bins	Bins	Bins	Bins
F(x)=				
1	525	1	1	3.33333
2	575	3	4	13.33333
3	625	4	8	26.66667
4	675	14	22	73.33333
5	725	4	26	86.66667
6	775	4	30	100
7				
8				



Tugas/Resume

- Cari struktur nanorod (cari gambar yang bagus). Cantumkan referensinya.
- Hitung rata2 **diameter** dan **panjangnya** menggunakan software.
- Data/statistik yang ditampilkan minimal 100 pengukuran (kecuali gambar tidak memungkinkan).
- Deadline: Minggu depan (sebelum kelas)